

Collaborative Teaching of Research Software Skills: Lessons from CodeRefinery for Higher Education and Research in Iceland

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Abstract (max 250 words)

Research software, data, and computational workflows are increasingly important across disciplines in higher education and research. At the same time, expectations related to reproducibility, transparency, open science, and FAIR research practices are becoming more prominent within Icelandic and international research environments. However, many students and researchers still receive limited formal training in collaborative coding, version control, testing, documentation, reproducibility, and sustainable research software practices. The aim of this work is to examine how collaborative teaching approaches inspired by the CodeRefinery project can support research software and open science education in Icelandic higher education and research environments.

The study draws on participation statistics from two international CodeRefinery workshops conducted during 2025 and 2026, together with teaching observations and analysis of anonymized participant questions and feedback collected throughout the workshops. The workshops employed a collaborative teaching model combining online instruction, collaborative exercises, local classroom support, peer learning, and openly available educational resources. The training focused on version control, collaborative software development, reproducible research workflows, software documentation, automated testing, and FAIR research software practices.

A total of 1,245 participants attended the two workshops, including 44 participants from Iceland. Participation increased from 505 participants in 2025 to 740 participants in 2026, while Icelandic participation increased from 17 to 27 participants. Analysis of participant questions and feedback revealed strong interest in practical topics such as version control, collaborative software development, reproducibility, testing, documentation, and computational notebooks. These findings indicate substantial demand for hands-on training in research software practices across disciplines. The collaborative teaching model also demonstrated the scalability of sharing expertise and educational resources across institutions and countries.

The study concludes that collaborative international training models provide an effective approach for strengthening research software competencies. Lessons learned highlight the value of combining local support with international expertise and suggest opportunities for integrating research software skills, reproducibility and open science practices into higher education curricula and professional development activities within the Icelandic community.

Keywords: research software engineering, collaborative teaching, reproducibility, open science, FAIR research software, higher education.